



BHABHA UNIVERSITY, BHOPAL

Scheme Of Examination

Third Semester- Master of Technology (Power Systems) Session 2019-20

SPECIALIZATION –POWER SYSTEM (PS)

S.No	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam	Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1.	EE-301	Elective- I	3	1	-	4	70	20	10	-	-	100
2.	EE-302	Elective- II	3	1	-	4	70	20	10	-	-	100
3.	EE-303	Seminar	-	-	4	4	-	-	-	-	100	100
4.	EE-304	Dissertation Part- I (Literature Review/Problem Formulation/ Synopsis)			8	8	-	-	-	120	80	200
		Total	6	2	12	20	140	40	20	120	180	500

Elective –I (EE- 301)

(A) Special Machine

(B) Advanced Electrical Drives

Elective-II (EE - 302)

(A) Power System Instrumentation

(B) DSP & its Application

(C) Power Controller



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF ENGINEERING**

Postgraduate Programme

(Courses effective from Academic Year 2019-20)

**SYLLABUS OF COURSES TO BE OFFERED
“Master of Technology”**

Disclaimer: The postgraduate programme in Master of Technology syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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Master of Technology
Branch - Electrical Engineering
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EE-301(A) SPECIAL MACHINE

UNIT-I

Stepper motor – variable reluctance stepper motor, single stack stepper motor multi stack stepper motor, permanent magnet stepper motor, Important features of stepper motor, torque v/s stepping rate characteristics, Drive circuits, unipolar drive circuits, Bipolar drive circuits.

UNIT-II

Switched reluctance motor, static torque production, partition of energy and the effects of saturation, Dynamic torque production, torque speed characteristics, shaft position sensing, solid rotors.

UNIT-III

Linear Induction Motors, construction, performance, thrust-speed characteristic, application, end effect.

UNIT-IV

Sine wave permanent magnet brushless dc motor, torque & emf equation, Inductance of phase winding, synchronous reactance, phasor diagram, torque-speed characteristics

UNIT-V

Square wave permanent magnet brushless dc motor, magnetic circuit analysis on open circuit torque & emf equations, torque speed characteristics, efficiency, commutation, winding inductances, armature reaction and controllers

REFERENCE BOOKS:-

1. Brushless Permanent Magnet & Reluctance Motor Drives – T.J.E.Miller
2. Principles of Electric Machines & Power Electronics – P.C.Sen
3. Electric Drives – G.K.Dubey

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EE-301(B) ADVANCE ELECTRICAL DRIVES

UNIT-I

Induction Motor Drives: Three phase I.M., analysis and performance. Operation with unbalanced source voltages and single phasing, analysis of I.M. fed from Non-sinusoidal voltage supply. Starting, Breaking, Introduction of transient analysis. Speed control methods, single phase I.M. Close loop control of I.M. Drives.

UNIT-II

Introduction of Brushless dc motor, stepper motor and switch reluctance motor drives, solar and battery powered drives, Traction Drives, Energy conservation in Electrical Drives.

UNIT-III

Synchronous Motor Drives, cylindrical rotor wound field motor, salient pole wound field motor, synchronous reluctance motor, Hysteresis synchronous motor, operation from fixed frequency supply, starting, breaking, synchronous motor variable speed drives, starting large synchronous machines.

UNIT-IV

Electrical Drives Introduction, Choice of Electrical Drives, Dynamics of Electrical Drives, Concept of Multi-quadrant operation, Components of load torques. Selection of motor power rating.

UNIT-V

D.C. Drive, speed torque, speed control. Starting, Breaking. Controlled rectified fed DC drive, chopper controlled dc drives. Close loop control of d.c. drive. Introduction of transient analysis

REFERENCE BOOKS:-

1. Power semi conductor controlled drives by -G.K.Dubey
2. Fundamentals of Electrical Drives by -G.K.Dubey
3. Electrical Machine & Power Electronics by- P.C.Sen

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EE-302(A) POWER SYSTEM INSTRUMENTATION

UNIT-I

Data transmission systems, Advantage and disadvantage of digital transmission over analog. Time division multiplexing, pulse modulation, digital modulation.

UNIT-II

Signal conditioning of inputs, single channel and multichannel data acquisition system, D/A and A/D converters, data loggers, supervisory control

UNIT-III

Transducers, sensors, actuators such as pressure, temperature, velocity, speed, volume, torque and solar flux measuring devices, current, voltage and power factor.

UNIT-IV

Introduction to instrumentation and control of energy systems, display instruments, recorders.

UNIT-V

Gas analysers, power plants and industrial instrumentation and pollution monitoring devices.

REFERENCE BOOKS:-

1. Transducers & Instrumentation by D.V.S. Murty – PHI Prentice Hall
2. Electronic Instrumentation by H.S.Kalsi – Tata McGraw Hill
3. Electrical and Electronics Measurement and Instr., A.K.Sawhney, Dhanpat Rai
4. Instrumentation devices and systems by C.S.Rangan and G.R. Sharma, TMH

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EE-302(B) DSP & ITS APPLICATION

UNIT-I

Introduction to DSP - Classification of signals, Multichannel and multi dimensional continuous v/s discrete time signals, continuous v/s discrete valued signals, continuous time sinusoidal signal, discrete time sinusoidal signals, sampling of analog signal, sampling theorem, quantification and coding of D/A conversion

UNIT-II

DSP Application - Introduction to digital signal processors chips, case study of different DSP applications. Application of filters to analog & digital signal processor, FET spectrum analyzer.

UNIT-III

Design Of Digital Filter - Design of linear phase FIR filter using window & frequency sampling method. Design of equiripple linear phase filters. Comparison of design methods for linear phase FIR filters. Design of IIR filters from analog filters. Direct Design Technique for digital IIR filters

UNIT-IV

Frequency Analysis Of Signals - Frequency analysis of analog signals, frequency analysis of discrete time signals. Properties of Fourier Transform, Frequency Domain Characteristics, Time Frequency Dualities, Sampling of signals in time and frequency domain, DFT & FFT.

UNIT-V

Discrete Time Signal and Systems - Discrete time signal, systems, Z-transform & Inverse Z-transform, analysis of discrete time, linear time invariant systems, co-relation of discrete time systems.

REFERENCE BOOKS:

1. Digital Signal Processing - W.D.Stanley
2. Analog & Digital Signal Processing – Ashok Ambardar

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EE-302(C) POWER CONTROLLER

UNIT-I

Various power semiconductor devices i.e. SCR, GTO, MOSFET, BJT, IGBT & MCT's & their protection, series-parallel operation, Heat sink calculations, Design of firing circuit for converters, choppers & inverters.

UNIT-II

1- ϕ to 1- ϕ , 3- ϕ to 3- ϕ cycloconverter circuits, circulating current scheme, non-circulating current operation, Mean output voltage, harmonics in supply current waveform & input-power factor. Concept of power quality

UNIT-III

Detailed analysis of 1- ϕ VSI, 3- ϕ VSI (180° mode, 150° mode & 120° mode of conduction), various inverter commutation circuits, harmonic reduction techniques, PWM inverters, Inverters for HVDC application & AC drives. Advantages & limitation of current source inverters over VSI, 1- ϕ and 3- ϕ CSI. Resonant inverters.

UNIT-IV

Analysis & design of voltage commutated, current commutated and load commutated choppers, multiquadrant choppers, and chopper for traction application. Resonant choppers, SMPS.

UNIT-V

Analysis & design of 1- ϕ bridge converter, 3- ϕ bridge converter with and without freewheeling diode, effect of source impedance, power factor improvement techniques, pulse width modulated converters, Dual converters, converter for HVDC application & DC drives.

REFERENCE BOOKS:

1. Thyristorised Power Controllers - G.K.Dubey, Doradla, Joshi, Sinha
2. Power Electronics - C.W.Lander
3. Power Electronics - Rashid
4. Thyristorised power controlled converters & cycloconverters - B.R.Pelly
5. Power Electronics – P S Bhimra